

VIKAS SHUKLA

Software Engineer ML • Machine Learning Systems • Full-Stack SWE

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Software Engineer specializing in building reliable, high-performance backend systems and data pipelines that scale. Proven ability to deliver production-ready solutions across cloud environments, turning complex engineering challenges into clean, maintainable software that drives measurable business outcomes.

EXPERIENCE

Software Engineer | Capital One | Richmond, VA

July 2023 – Present

- Architected and shipped a hybrid ML plus rule-based software pipeline using XGBoost for real-time account compromise detection; reduced fraud analyst investigation workload by 45 hours/month and improved model precision by 17%.
- Engineered automated Python/SQL data quality monitoring service with real-time KPI alerting integrated into the CI/CD pipeline, intercepted data issues pre-production in ~80% of cases and reduced weekly metric variance by 20%.
- Designed and built a scalable data ingestion and transformation layer consolidating 40M+ transactions from disparate systems into a unified self-serve analytics platform, eliminating 60% of manual analysis cycles.
- Developed and productionized fraud scoring microservices with automated model threshold management via CI/CD, achieved 98% model adoption and zero-downtime deployments across Risk Management teams.
- Collaborated with engineering, compliance, and legal teams to align ML model outputs with U.S. regulatory standards; implemented monitoring dashboards, model diagnostics, and root-cause analysis tooling.

Software Engineer ML | Amazon | Bellevue, WA

Jun 2022 – Apr 2023

- Designed and deployed secure notebook infrastructure for ML model training and deployment, reducing model development-to-production time by 40% for the Alexa Data Science team.
- Built AI-powered notification quality assessment framework integrating user settings, reachability signals, and A/B test results across push, SMS, and email channels; improved Alexa user engagement and retention by 20%.
- Launched uplift-based targeting system with guardrails against notification fatigue: reduced total sends by ~22% with no engagement loss, cut SMS/email failures by ~30%, and lifted incremental conversion by ~8%.
- Implemented automated business logic for A/B, multivariate, and auto-targeting marketing experiments using Python, increased campaign ROI and strategy effectiveness.

Software Engineer | SGS Teknics | Gurugram, India

Nov 2014 – May 2018

- Built and maintained REST API integrations to pull data from third-party vendor systems into the internal dashboards, tested endpoints, handled error responses, and documented API contracts for the team.
- Developed and maintained internal CRUD application features using python and JavaScript to manage product catalog and customer records, fixed bugs, wrote unit tests, and participated in code reviews.
- Built demand forecasting module using regression models in Python integrated into the order management system, expand results via RESTAPI endpoint consumed by the frontend dashboard, reducing overstock incidents by ~15%

KEY PROJECT

IntelliQuery Engine (LLM + RAG Agent) [🔗](#) | Python, FastAPI, FAISS, React Native, SQL

- Built retrieval-augmented generation system using LLM embeddings and FAISS vector search enabling semantic queries across 10K+ document chunks with <800ms response time; reduced unsupported responses by ~60%.
- Implemented intent-routing with lightweight classification for finance/legal queries, improving accuracy by ~45% and cutting manual validation effort by ~50%.
- Deployed scalable FastAPI backend integrated with React Native application ([caseFit](#)) delivering production-grade low-latency AI responses.

EDUCATION

M.S Computer & Information Studies | Trine University, Reston, VA

Feb 2025

M.S. Data Science (GPA: 3.7 / 4.0) | University at Buffalo

Feb 2022

B.Tech. Electrical & Electronics Engineering (GPA: 8.6 / 10) | SRM University, India

May 2014

TECHNICAL SKILLS

Languages: Python, Java, C#, JavaScript, C++, SQL, React JS, React Native, TypeScript, FastAPI, REST APIs, HTML / CSS, Node.js, Data Structure and Algorithms, Linux, OOPs

Cloud & DevOps: AWS Services (Lambda, S3, ECS Fargate, DynamoDB, RDS, SQS, SNS, Kinesis, CloudFormation, Glue Event Bridge, AppConfig), Azure Services (CosmosDB, Kafka, Kubernetes)